Read More

- Read 1/5 - https://medium.com/@iot24hours/top-7-iot-protocols-16e1cbbdaff0
- Read 2/5 - https://medium.com/@iot24hours/ทำความรู้จักกับ-mqtt-protocol-สำหรับระบบ-iot-ที่จำเป็นต้องรู้-9508957a8b61
- Read 3/5 - https://medium.com/@iot24hours/7-เทคโนโลยีทางด้าน-iot-ที่เหมาะสมกับการปรับใช้ในประเทศไทย-b75f8a9920eb
- Read 4/5 - https://medium.com/@iot24hours/10-แนวโน้มของการเปลี่ยนแปลงด้าน-iot-internet-of-thing-ที่อาจเกิดขึ้นในปี-2020-c26684d7e6da
- Read 5/5 - https://medium.com/@iot24hours/testing-our-django-app-to-be-publicly-accessible-using-ngrok-9b73851c46e0

Experiment

1. Install Node-RED and Start
2. Mission-1: Creating your first flow
3. Mission-2: Node-RED Publish and Subscribe
4. Mission-3: LINE Test >> https://www.youtube.com/watch?v=2vGtXZRlGy8
5. Mission-4: LINE Bot with Node-RED >> https://youtu.be/4KKIEmWYCGQ?t=794
6. Mission-5: From ESP-32 to MQTT Broker and If xxx then LINE Alert
    - หากอุณหภูมิ เกิน xxxx หรือ ความชื้น เกิน yyyy ให้มีการแจ้งเตือนผ่าน LINE
7. Mission-6: From ESP-32 to MQTT Broker and Show on Node-RED Dashboard
    - แสดงอุณหภูมิและความชื้นจาก DHT-22 ที่ Node-RED Dashboard
    - ควบคุม LED 2 ตัว

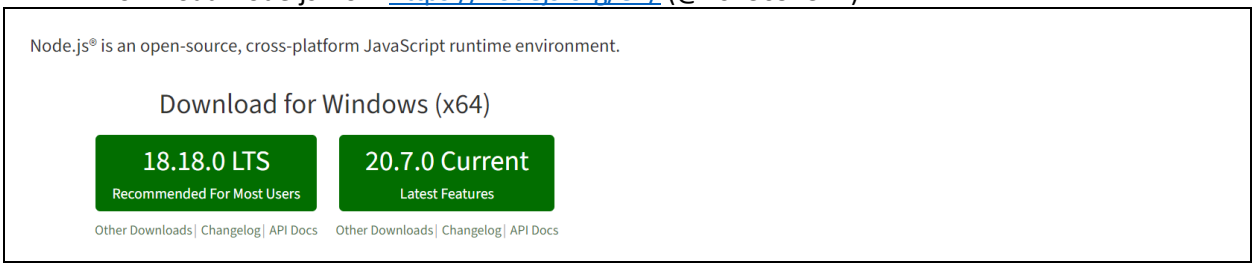
Week0920.20230925 – Node-RED

Read More

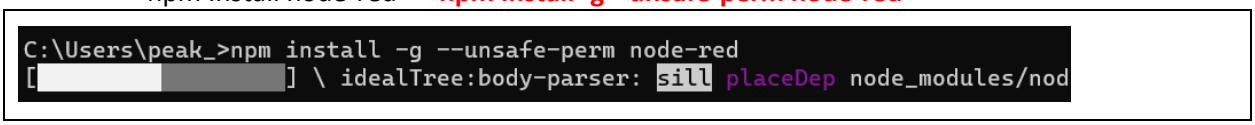
- http://noderedguide.com/
- Install and First Flows
  - https://nodered.org/docs/tutorials/first-flow
  - https://www.youtube.com/watch?v=jqJ-D41SFL8
  - https://www.youtube.com/watch?v=rQuyjF2_Q4M
- Install https://jackrobotics.me/node-red-ep0-install-windows-1c836a715fe
- Get LINE Token https://jackrobotics.me/แจ้งเตือน-line-notify-เตือนง่ายได้ใจ-7328125083ec
- Data to LINE with node-red https://jackrobotics.me/จับ-line-notify-ยัด-node-red-โดยใช้ความสามารถเดิมของ-node-red-1730d83c067c

Install

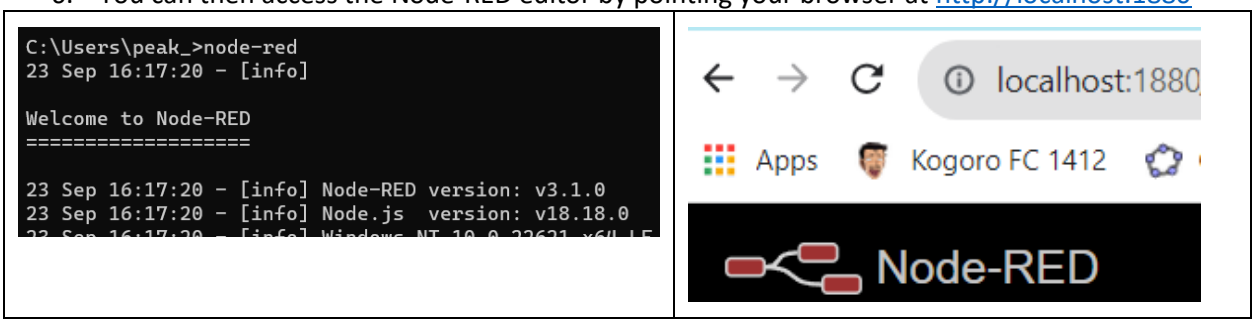
1. Getting Start https://nodered.org/docs/getting-started/
2. Download Node.js from https://nodejs.org/en/ (@20230923>>)



3. Install Node.js >> node-v18.18.0-x64.exe
4. Install node-red https://nodered.org/docs/getting-started/windows
   - Start → nodejs command prompt
   - npm Install node-red >> **npm install -g --unsafe-perm node-red**



5. Start node-red >> **node-red**
6. You can then access the Node-RED editor by pointing your browser at http://localhost:1880



Mission-1: Creating your first flow

Overview

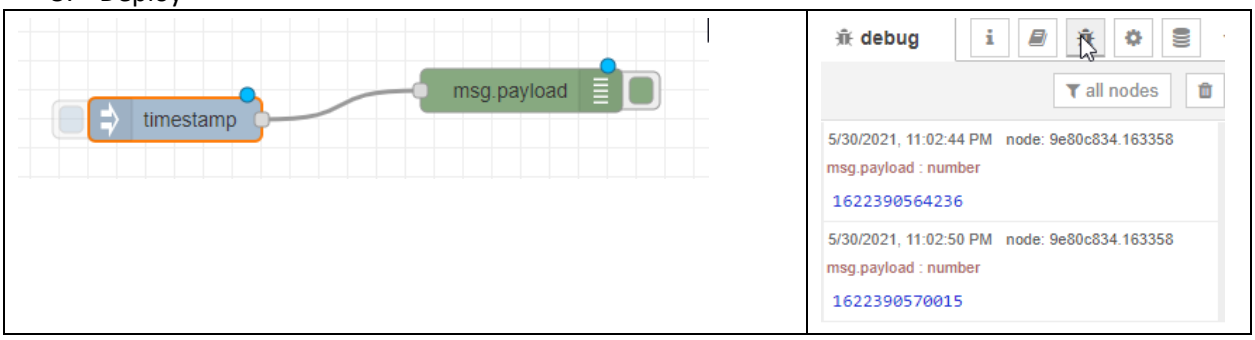
This tutorial introduces the Node-RED editor and creates a flow that demonstrates the Inject, Debug and Function nodes.

1. Access the editor
   - Start → nodejs command prompt
   - พิมพ์ node-red
   - Run chrome with url → http://localhost:1880

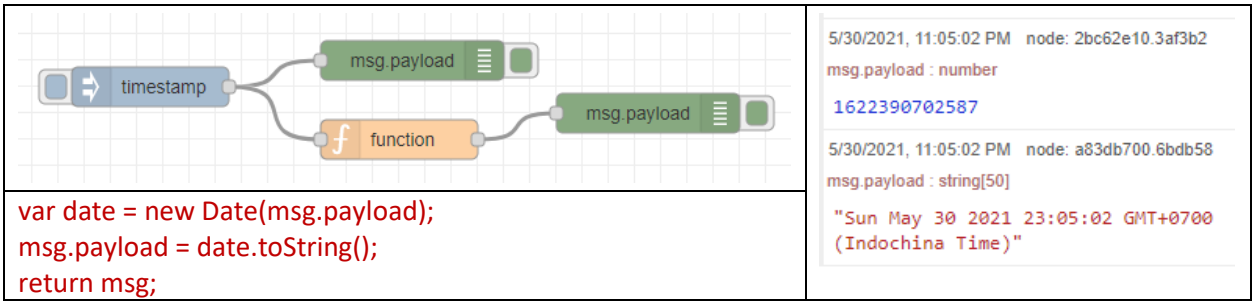
```
Your environment has been set up for using Node.js 14.17.0 (x64) and npm.

C:\Users\Pk007_20210404>node-red
```

2. Add an Inject node
3. Add a Debug node
4. Wire the two together
5. Deploy



6. Add a Function node

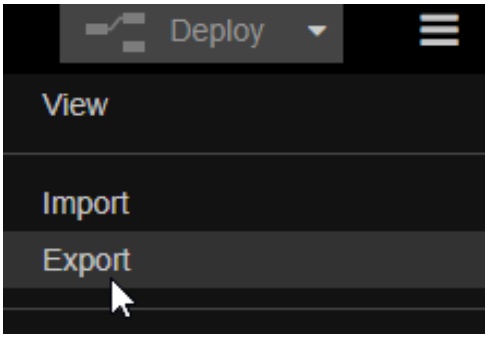


Summary

This flow demonstrates the basic concept of creating a flow. It shows how the Inject node can be used to manually trigger a flow, and how the Debug node displays messages in the sidebar. It also shows how the Function node can be used to write custom JavaScript to run against messages.

Source

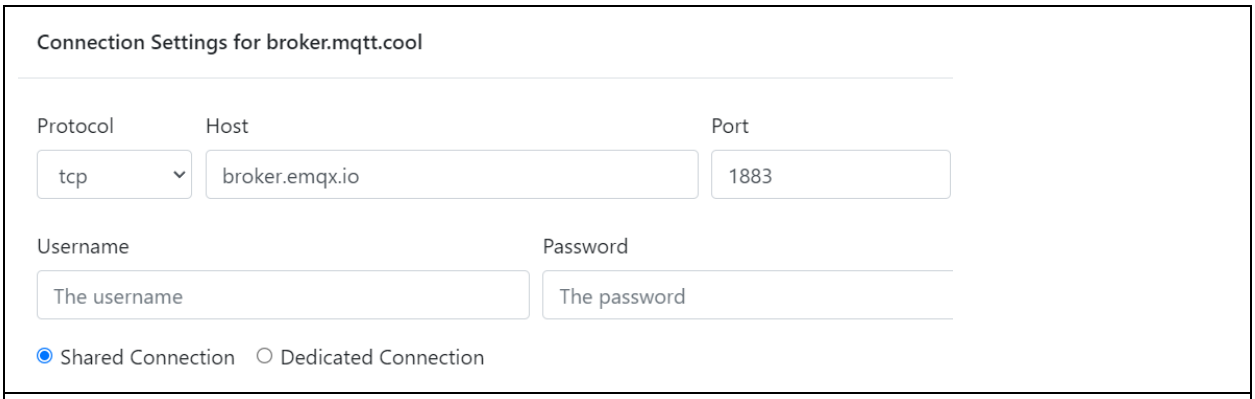
The flow created in this tutorial is represented by the following json. To import it into the editor, copy it to your clipboard and then paste it into the Import dialog.

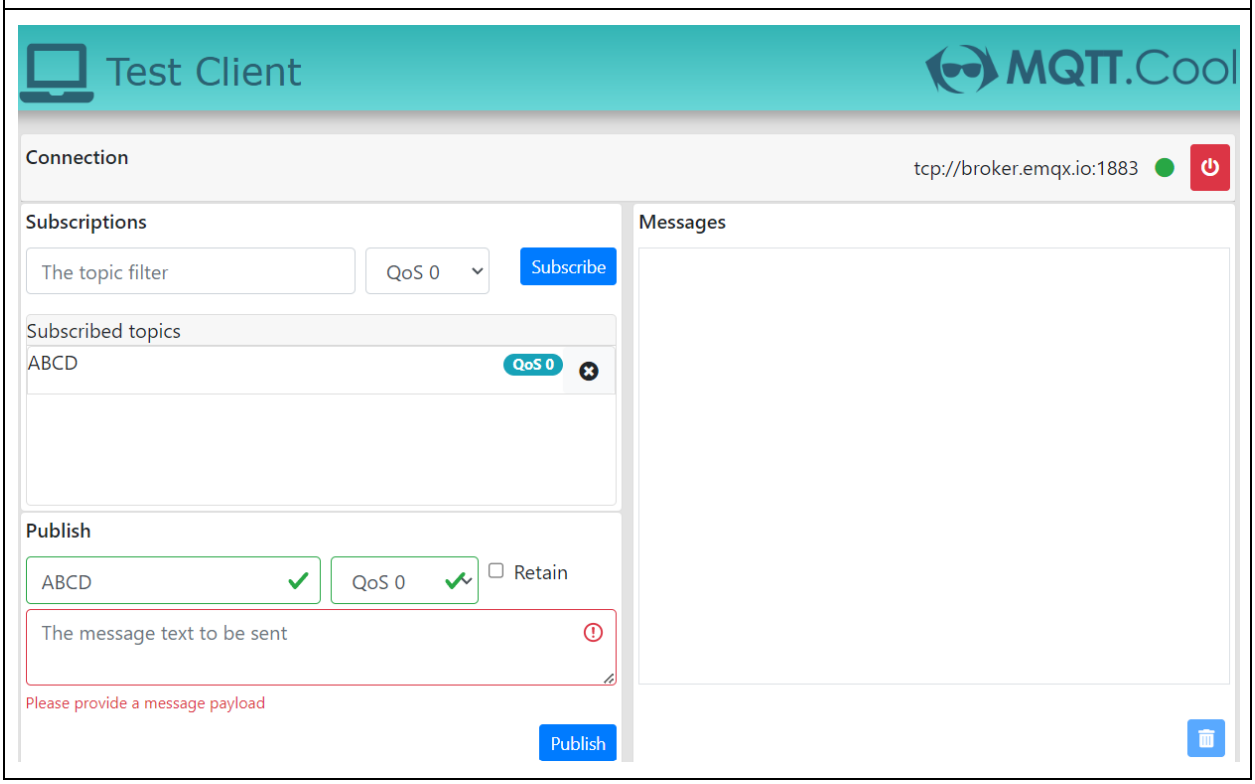


[{"id":"7b11139d.49fbbc","type":"tab","label":"Flow 1","disabled":false,"info":""},{"id":"189da234.1c1bce","type":"inject","z":"7b11139d.49fbbc","name":"","props":[{"p":"payload"},{"p":"topic","vt":"str"}],"repeat":"","crontab":"","once":false,"onceDelay":0.1,"topic":"","payload":"","payloadType":"date","x":190,"y":140,"wires":[["2bc62e10.3af3b2","9d9bf503.b23838"]]},{"id":"2bc62e10.3af3b2","type":"debug","z":"7b11139d.49fbbc","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":390,"y":120,"wires":[]},{"id":"9d9bf503.b23838","type":"function","z":"7b11139d.49fbbc","name":"","func":"// Create a Date object from the payload\nvar date = new Date(msg.payload);\n// Change the payload to be a formatted Date string\nmsg.payload = date.toString();\n// Return the message so it can be sent on\nreturn msg;","outputs":1,"noerr":0,"initialize":"","finalize":"","libs":[],"x":380,"y":180,"wires":[["a83db700.6bdb58"]]},{"id":"a83db700.6bdb58","type":"debug","z":"7b11139d.49fbbc","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":570,"y":160,"wires":[]}]

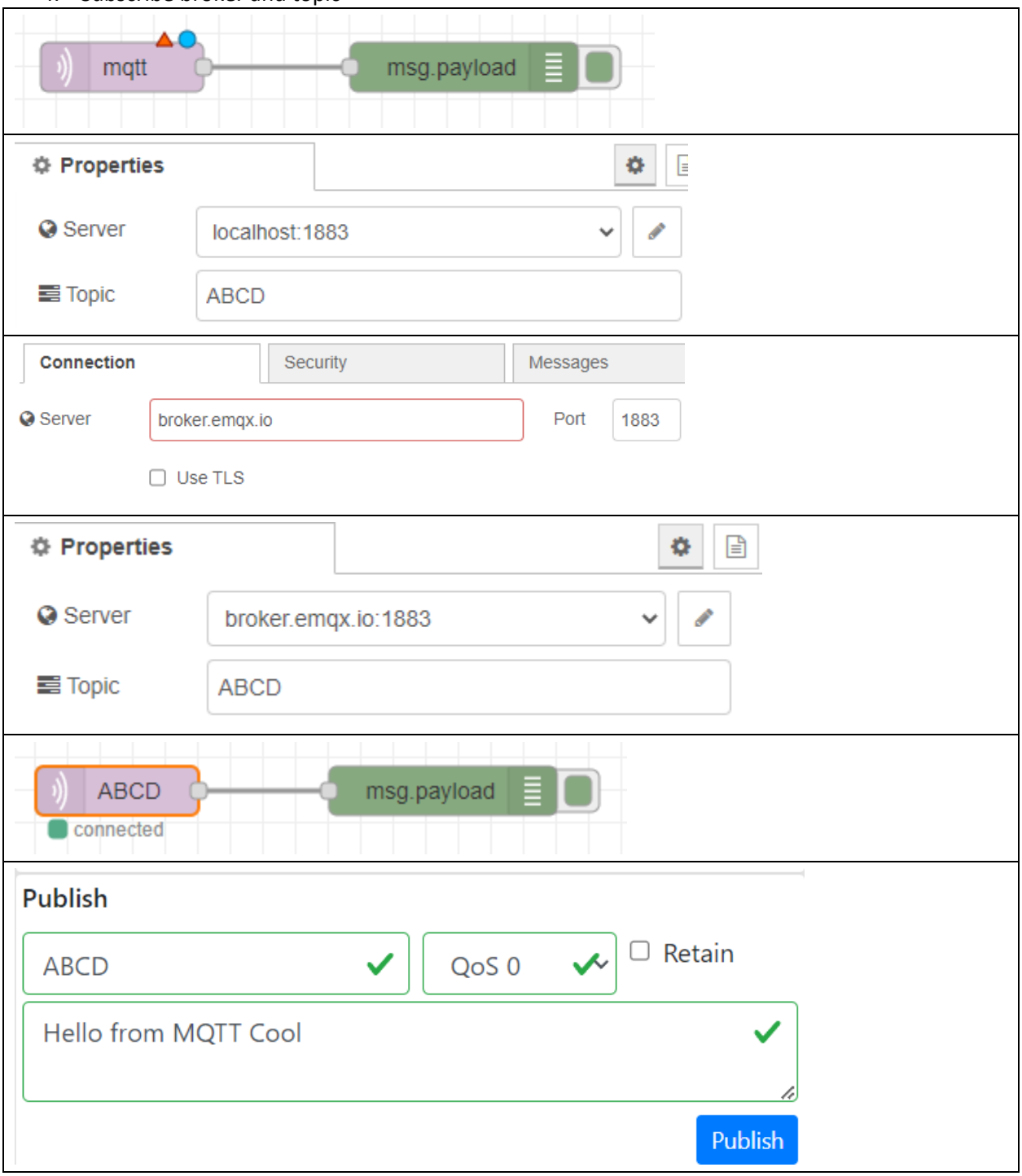
Mission-2: Publish and Subscribe

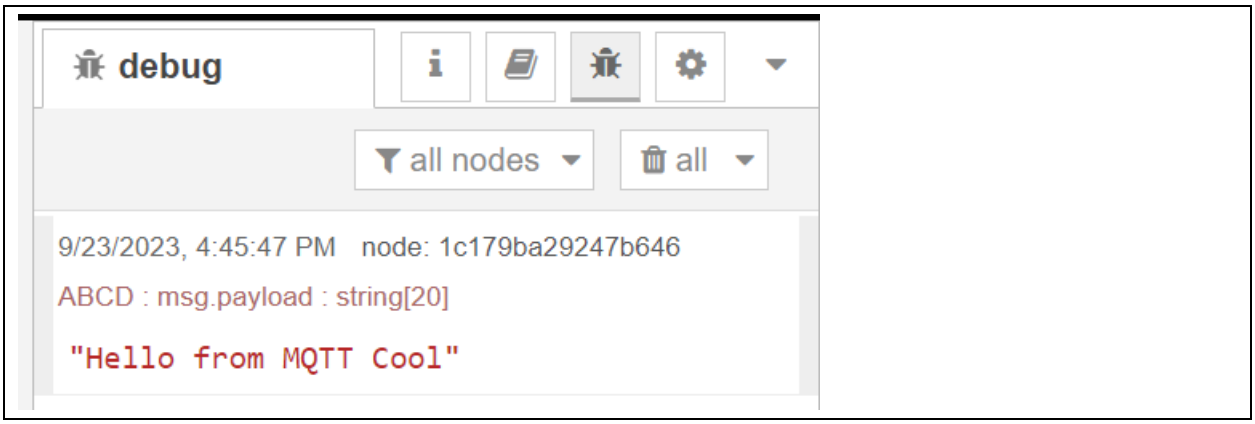
1. Open MQTT Test Client >> https://testclient-cloud.mqtt.cool/
2. Select Free mqtt broker >> **Broker: broker.emqx.io, TCP Port: 1883**
3. Select Topic "ABCD" for Subscribe and Publish



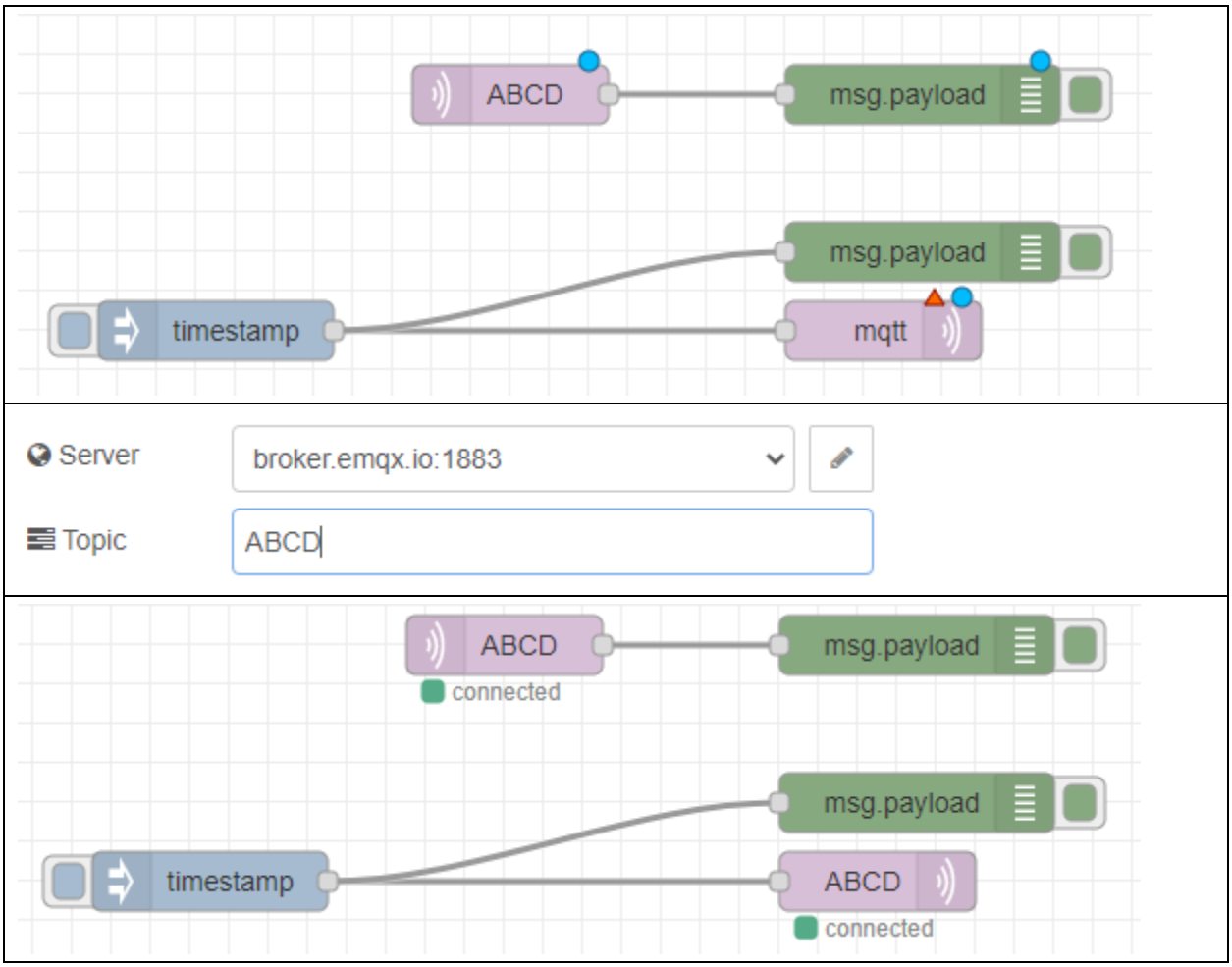


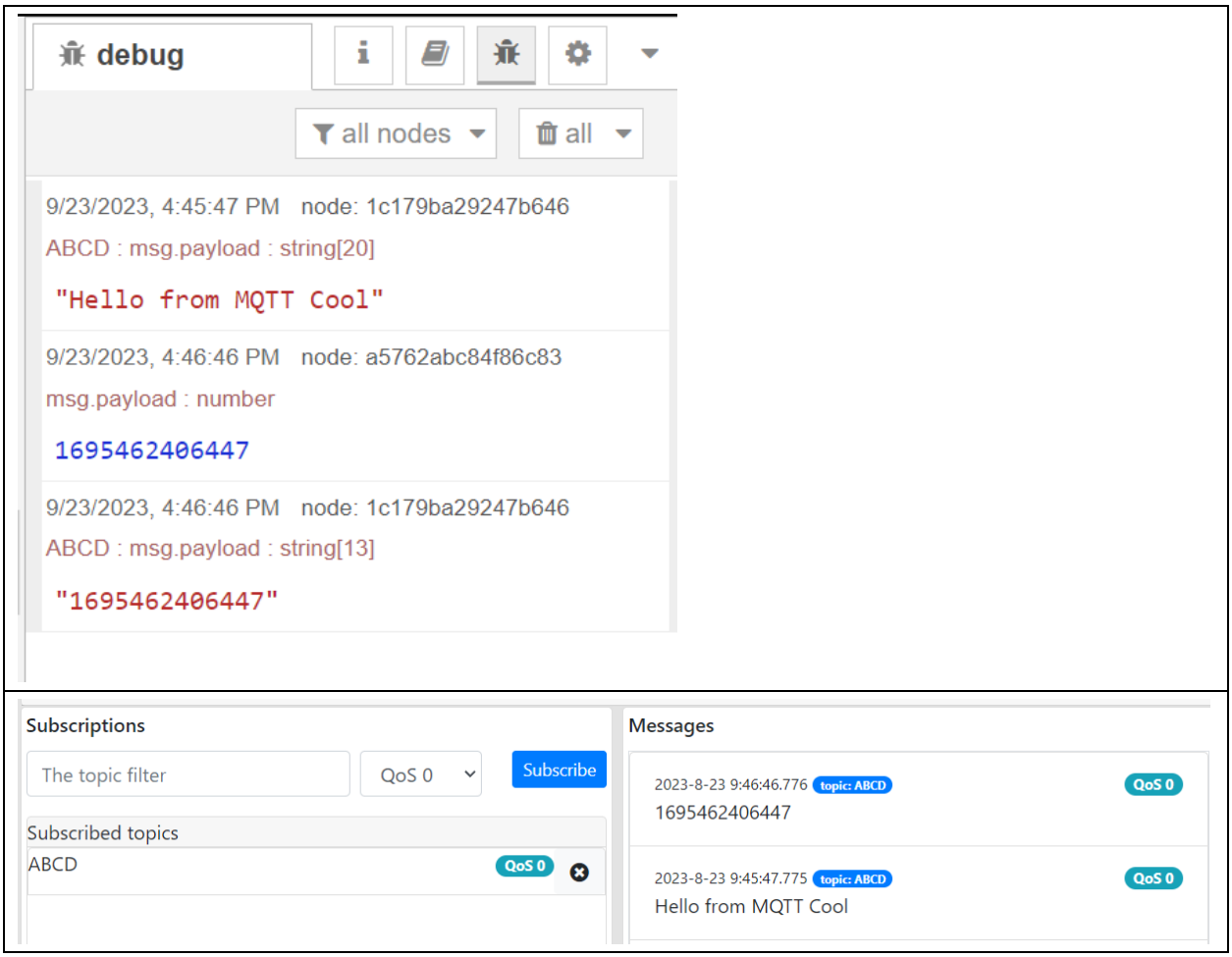
4. Subscribe broker and topic




debug
all nodes
all
9/23/2023, 4:45:47 PM node: 1c179ba29247b646
ABCD : msg.payload : string[20]
"Hello from MQTT Cool"

5. Publish


ABCD
msg.payload
msg.payload
timestamp
mqtt
Server
broker.emqx.io:1883
Topic
ABCD
ABCD
connected
msg.payload
msg.payload
timestamp
ABCD
connected



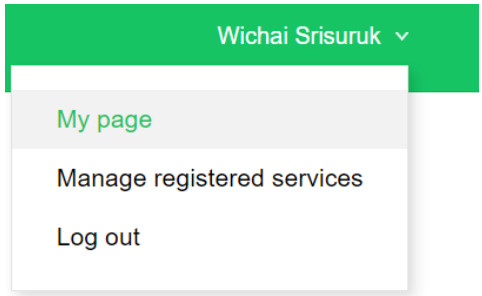
6. Source

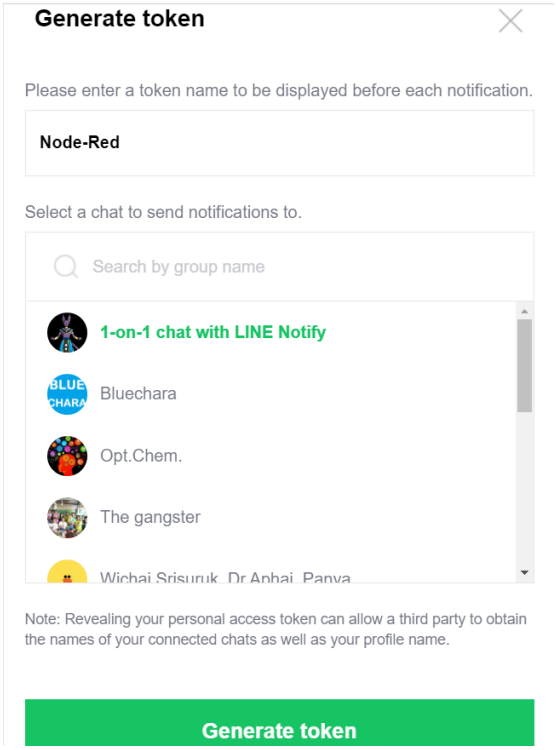
[{"id":"7b11139d.49fbbc","type":"tab","label":"Flow 1","disabled":false,"info":""},{"id":"2bc62e10.3af3b2","type":"debug","z":"7b11139d.49fbbc","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":550,"y":240,"wires":[]},{"id":"a83db700.6bdb58","type":"debug","z":"7b11139d.49fbbc","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":550,"y":320,"wires":[]},{"id":"76082819.de1288","type":"mqtt out","z":"7b11139d.49fbbc","name":"","topic":"ABCD","qos":"","retain":"","respTopic":"","contentType":"","userProps":"","correl":"","expiry":"","broker":"ef016747.18fa38","x":530,"y":360,"wires":[]},{"id":"9f4c6034.7deda","type":"inject","z":"7b11139d.49fbbc","name":"","props":[{"p":"payload"},{"p":"topic","vt":"str"}],"repeat":"","crontab":"","once":false,"onceDelay":0.1,"topic":"","payload":"","payloadType":"date","x":190,"y":360,"wires":[["76082819.de1288","a83db700.6bdb58"]]},{"id":"fecf4aca.eb9ae8","type":"mqtt in","z":"7b11139d.49fbbc","name":"","topic":"ABCD","qos":"2","datatype":"auto","broker":"ef016747.18fa38","nl":false,"rap":true,"rh":0,"x":340,"y":240,"wires":[["2bc62e10.3af3b2"]]},{"id":"ef016747.18fa38","type":"mqtt-broker","name":"","broker":"broker.emqx.io","port":"1883","clientid":"","usetls":false,"protocolVersion":"4","keepalive":"60","cleansession":true,"birthTopic":"","birthQos":"0","birthPayload":"","birthMsg":{},"closeTopic":"","closeQos":"0","closePayload":"","closeMsg":{},"willTopic":"","willQos":"0","willPayload":"","willMsg":{},"sessionExpiry":""}]

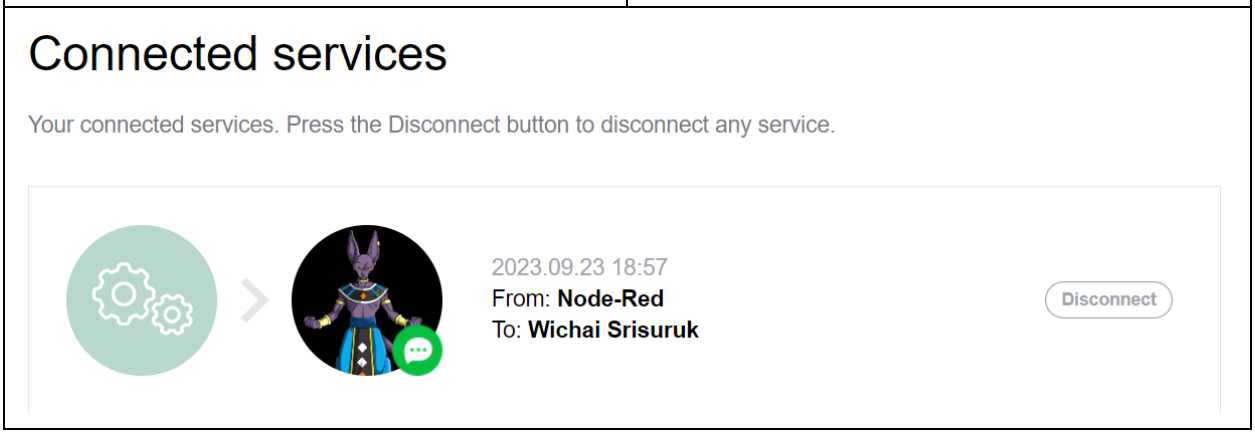
Mission-3: LINE Test

→ **Follow This >> https://www.youtube.com/watch?v=2vGtXZRlGy8**

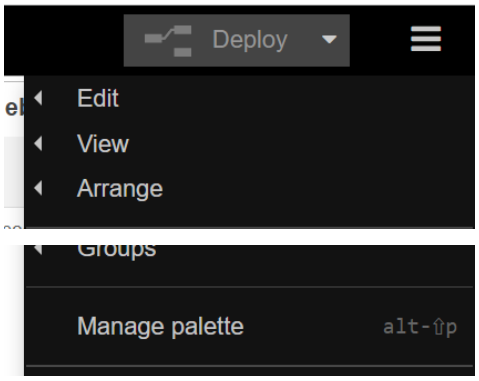
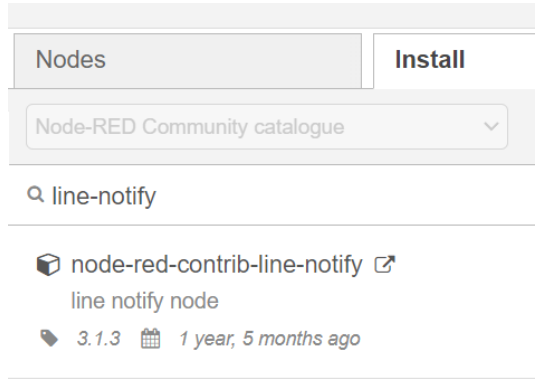
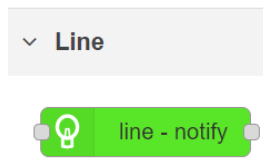
1. Get LIN Token
   1.1 https://notify-bot.line.me/en/
   1.2 Login, My Page, Generate access token (For developers)



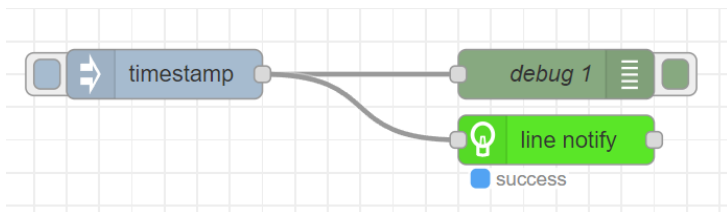
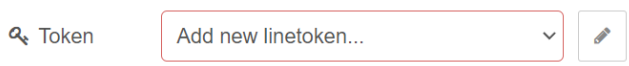
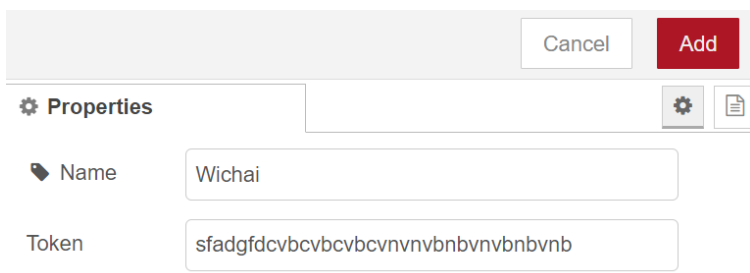




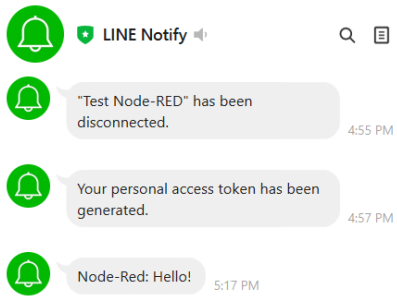
2. Add LINE Node
   2.1 Manage Palate
   2.2 Filter by “line-notify”, Install

3. Inset node to flow and config

Add Node • timestamp • debug • line notify	
Line notify 1. Add New token	
2. Fill Token 3. Add	
4. Using Token	

4. Deploy and Test

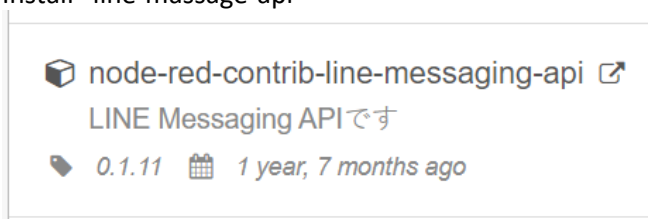


5. Source

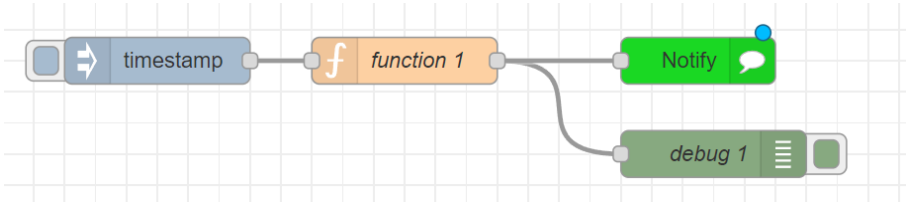
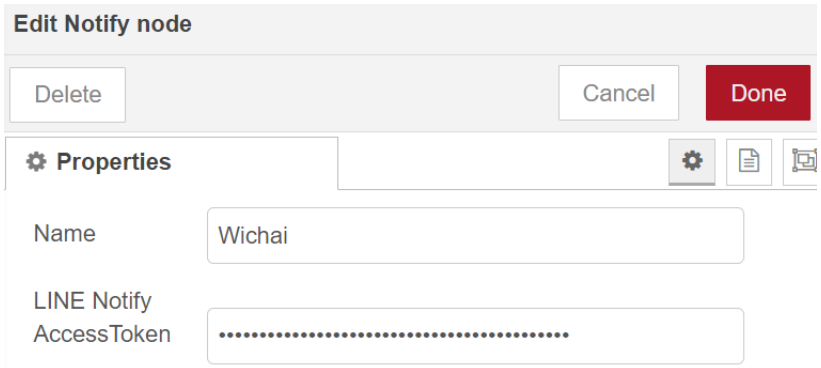
[{"id":"9a221616b050daf7","type":"tab","label":"Flow 1","disabled":false,"info":""},{"id":"032c6c386cf0e4cf","type":"inject","z":"9a221616b050daf7","name":"","props":[{"p":"payload"},{"p":"topic","vt":"str"}],"repeat":"","crontab":"","once":false,"onceDelay":0.1,"topic":"","payload":"","payloadType":"date","x":140,"y":160,"wires":[["f9707767fb9592d4","c59006bf2710c0a9"]]},{"id":"f9707767fb9592d4","type":"debug","z":"9a221616b050daf7","name":"debug 1","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":380,"y":160,"wires":[]},{"id":"c59006bf2710c0a9","type":"line-notify","z":"9a221616b050daf7","name":"","message":"Hello!","contentType":"message","imageThumbnail":"","imageUrl":"","sticker":"default","stickerPackageId":"1","stickerId":"1","silent":false,"creds":"5f67b39057b8b538","x":380,"y":220,"wires":[[]]},{"id":"5f67b39057b8b538","type":"linetoken","name":"Wichai"}]

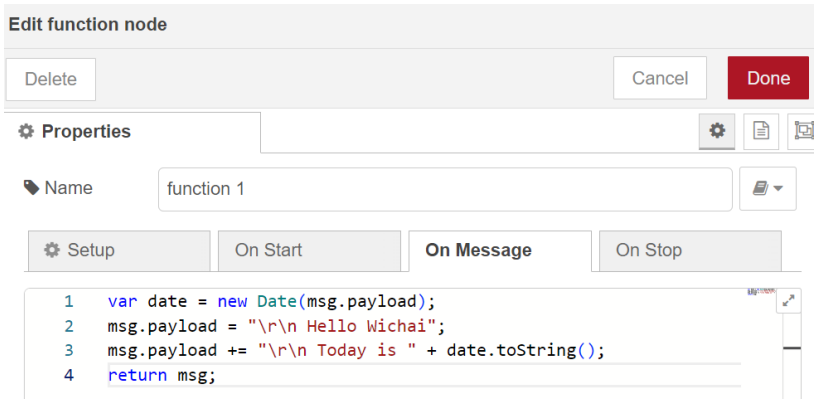
6. Test with LINE Massage API

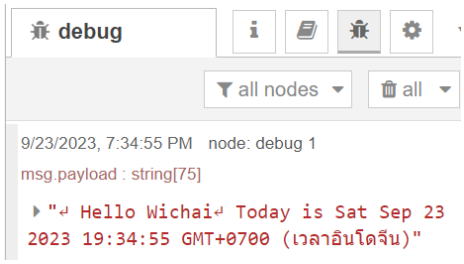
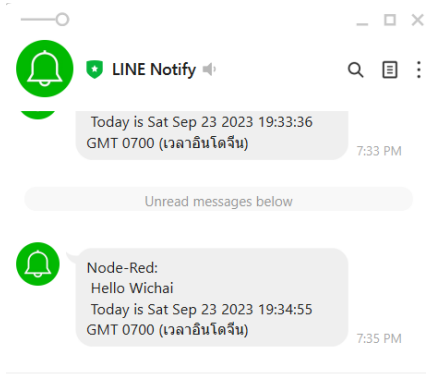
- Manage Pallets
- Install “line-massage-api”



7. Test This

Add Node • Timestamp • Function • Notify • debug	
Edit Notify • Name • LINE Token	

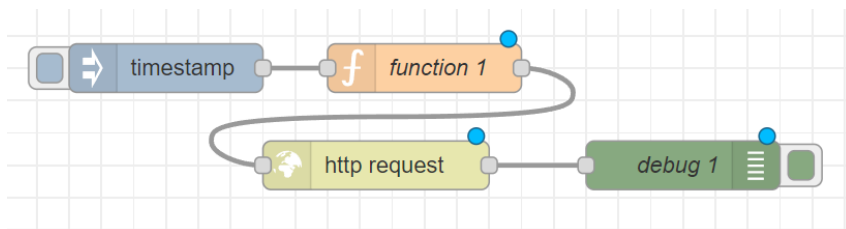
Edit function	```
	var date = new Date(msg.payload);
	msg.payload = "\r\n Hello Wichai";
	msg.payload += "\r\n Today is " + date.toString();
	return msg;
	``
	 

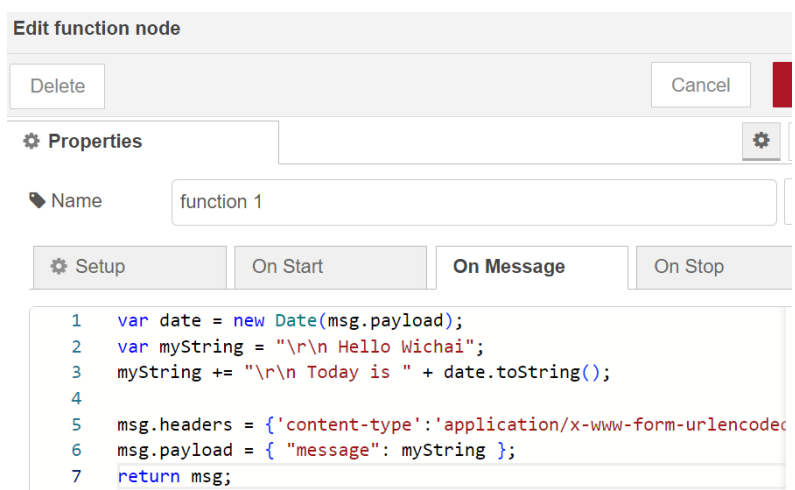
Deploy & Test	 

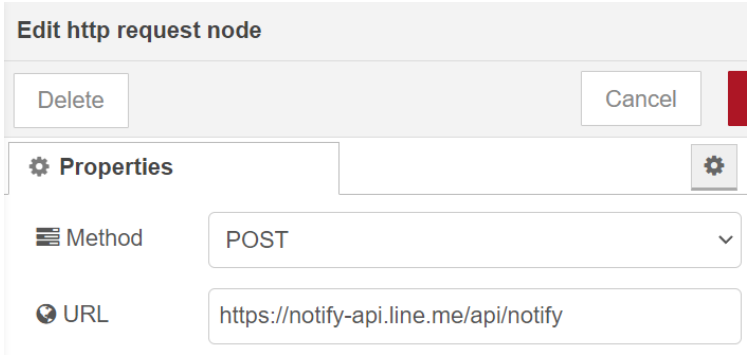
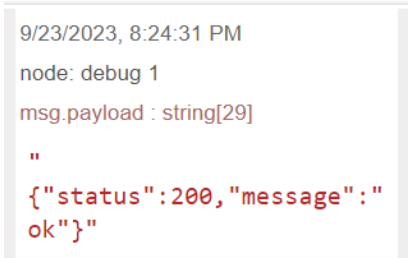
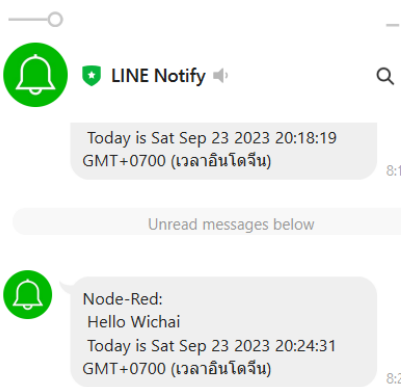
8. Source Code

[{"id":"9a221616b050daf7","type":"tab","label":"Flow 1","disabled":false,"info":""},{"id":"032c6c386cf0e4cf","type":"inject","z":"9a221616b050daf7","name":"","props":[{"p":"payload"},{"p":"topic","vt":"str"}],"repeat":"","crontab":"","once":false,"onceDelay":0.1,"topic":"","payload":"","payloadType":"date","x":140,"y":160,"wires":[["595edaaeedc81eec"]]},{"id":"f9707767fb9592d4","type":"debug","z":"9a221616b050daf7","name":"debug 1","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":500,"y":160,"wires":[]},{"id":"b8b9e2fb0cb8013d","type":"Notify","z":"9a221616b050daf7","name":"Wichai","x":490,"y":200,"wires":[]},{"id":"595edaaeedc81eec","type":"function","z":"9a221616b050daf7","name":"function 1","func":"var date = new Date(msg.payload);\nmsg.payload = \"\\r\\n Hello Wichai\";\nmsg.payload += \"\\r\\n Today is \" + date.toString();\nreturn msg;","outputs":1,"timeout":0,"noerr":0,"initialize":"","finalize":"","libs":[],"x":320,"y":160,"wires":[["f9707767fb9592d4","b8b9e2fb0cb8013d"]]}]

9. Test with https request < https://www.youtube.com/watch?v=WE3rESOqJeM >

Add Node • Timestamp • Function • http request • debug	

Edit function	`var date = new Date(msg.payload);`
	`var myString = "\r\n Hello Wichai";`
	`myString += "\r\n Today is " + date.toString();`
	
	`msg.headers = {'content-type':'application/x-www-form-urlencoded','Authorization':'Bearer LINE_Token'};`
	`msg.payload = { "message": myString };`
	`return msg;`
	 

Edit http request	Method = POST URL = https://notify-api.line.me/api/notify 	
Deploy & Test		

10. Source Code


```
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```


11. จาก https://medium.com/@nattaponsirikamonnet/มาลอง-line-notify-กันเถอะ-sticker-a2d25925d1a1 >ลองปรับค่าของ msg.payload .ใน function ให้เป็นดังนี้ “`msg.payload = {"message": myString,"stickerPackageId":1,"stickerId":2};`”

12. จาก https://www.youtube.com/watch?v=TyNyoMRLJ9Y ต้องการส่งรูปภาพไปที่ LINE ด้วย Node-RED ต้องทำอย่างไรบ้าง

Mission-4: LINE Chat

→ **Follow This >> https://youtu.be/4KKIEmWYCGQ?t=794**

1. Xxxx

2. Yyyyy

3. Source

< ข้ามไปก่อน สะดวกค่อยกลับมาทำ>

Mission-5: From ESP32 to MQTT Broker and If xxx Then LINE Alert

1. System
2. ESP-32 Publish to MQTT Broker
3. node-red on PC Subscribe Broker
4. node-red code if Temperature >25 LINE Alert

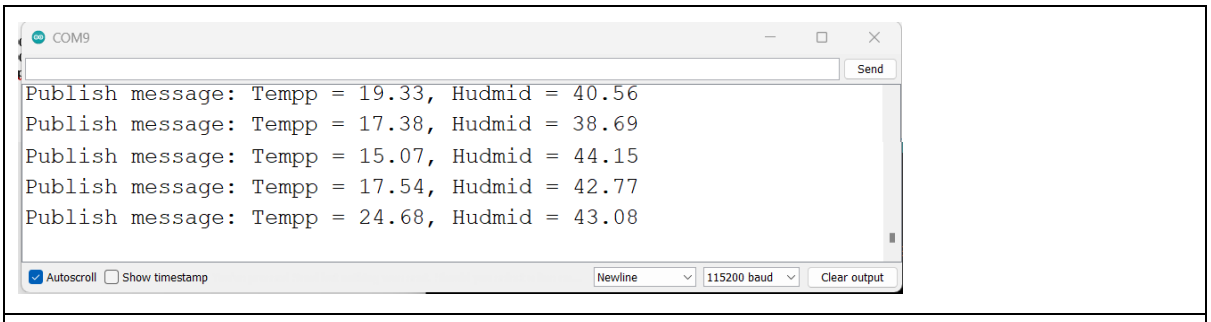
   …..

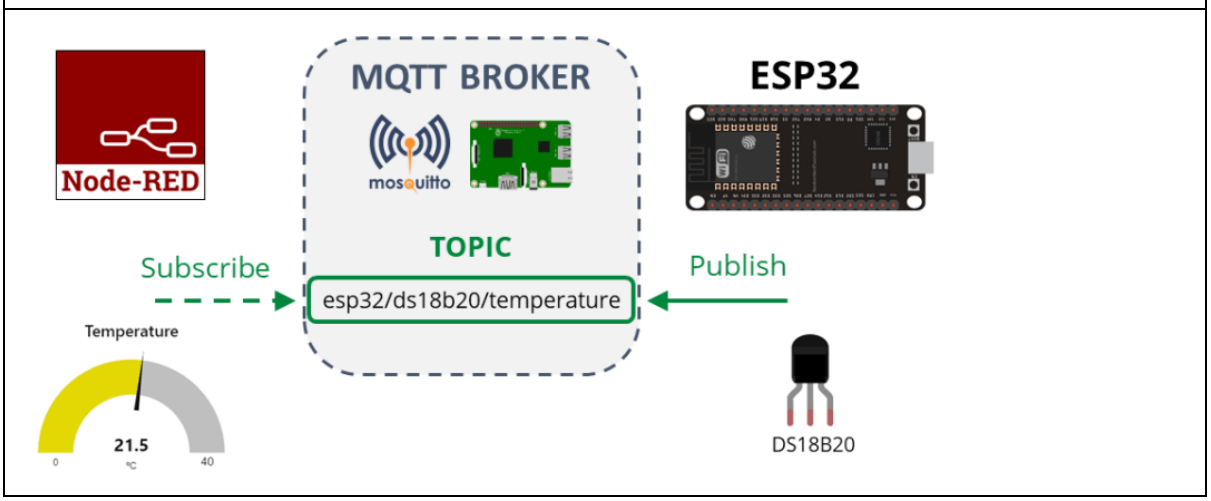
5. source

Mission-6: From ESP32 to MQTT Broker and Show on NodeRED Dashboard

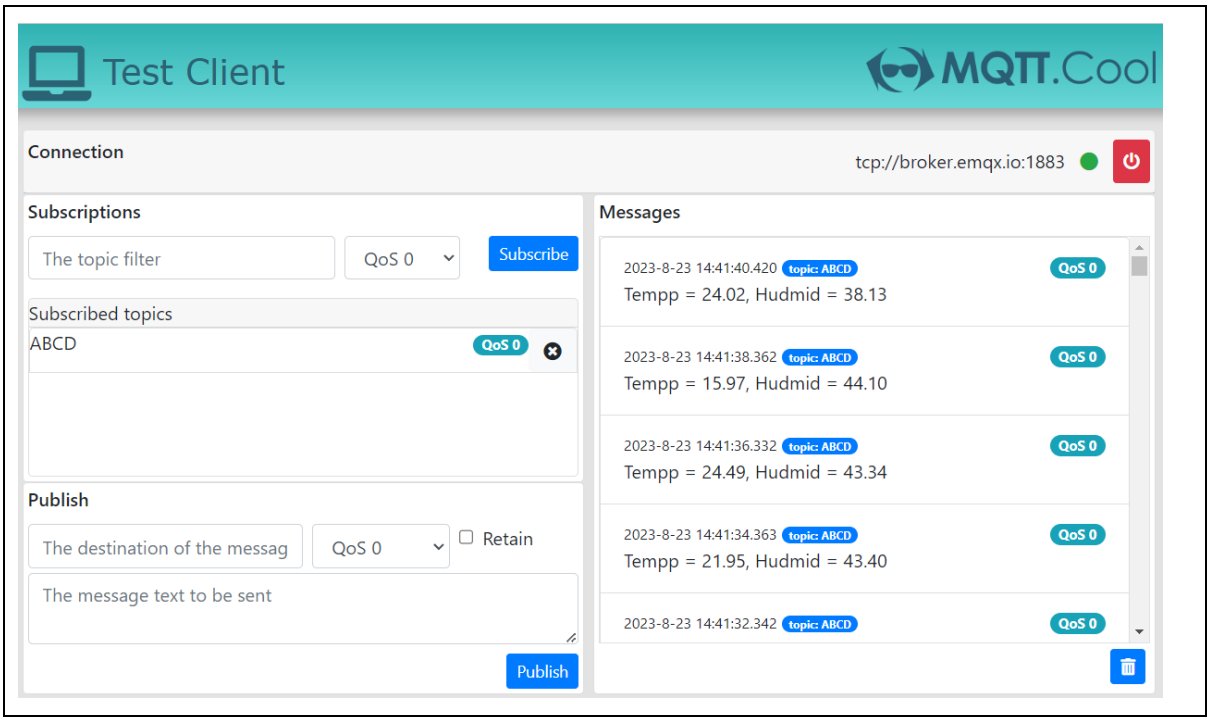
1. Load ESP-32

```
#include <WiFi.h>
#include <PubSubClient.h>

const char* ssid = "Test1234";
const char* password = "0816601929";
const char* mqtt_server = "broker.emqx.io";
const char* myTopic = "ABCD";

#define MSG_BUFFER_SIZE  (50)

WiFiClient espClient;
PubSubClient client(espClient);
unsigned long lastMsg = 0;
char msg[MSG_BUFFER_SIZE];
int value = 0;

void setup_wifi() {
  delay(10);
  Serial.print("/nConnecting to ");            Serial.println(ssid);
  WiFi.mode(WIFI_STA);                          WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  randomSeed(micros());
  Serial.println("");                           Serial.println("WiFi connected");
  Serial.println("IP address: ");               Serial.println(WiFi.localIP());
}

void reconnect() {
  while (!client.connected()) {
    Serial.print("Attempting MQTT connection...");
    String clientId = "ESP32 Client-";
    clientId += String(random(0xffff), HEX);
    if (client.connect(clientId.c_str())) {
      Serial.println("connected");
      client.subscribe(myTopic);
      client.publish(myTopic, "hello world");
    } else {
      Serial.print("failed, rc=");
      Serial.print(client.state());
      Serial.println(" try again in 5 seconds");
      delay(5000);
    }
  }
}

void setup() {
  Serial.begin(115200);
  setup_wifi();
  client.setServer(mqtt_server, 1883);
}

void loop() {
  if (!client.connected())     reconnect();
  client.loop();
  unsigned long now = millis();
  if (now - lastMsg > 2000) {
    float value1 = 15 + random(0, 1000) / 100.0;
    float value2 = 35 + random(0, 1000) / 100.0;
    snprintf (msg, MSG_BUFFER_SIZE, "Tempp = %0.2f, Humid = %0.2f", value1, value2);
    Serial.print("Publish message: ");
    Serial.println(msg);
    client.publish(myTopic, msg);
    lastMsg = millis();
  }
}
```

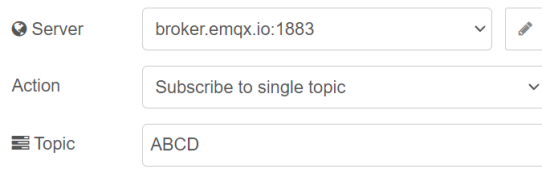


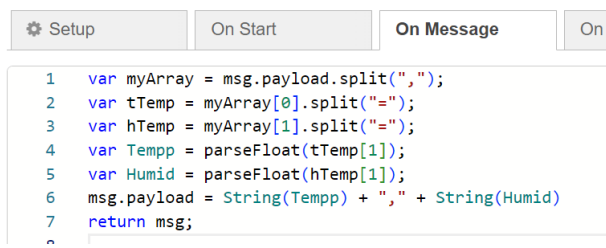


2. Monitor with MQTT.Cool
   - Broker = broker.emqx.io:1883
   - Topic = ABCD



3. Read from Broker

Add Node • Mqtt in • Function • debug	
Edit Mqtt in • Broker = broker.emqx.io:1883 • Topic = ABCD	

Edit Function	`var myArray = msg.payload.split(",");`
	`var tTemp = myArray[0].split("=");`
	`var hTemp = myArray[1].split("=");`
	`var Tempp = parseFloat(tTemp[1]);`
	`var Humid = parseFloat(hTemp[1]);`
	`msg.payload = String(Tempp) + "," + String(Humid)`
	`return msg;`
	 

Deploy	

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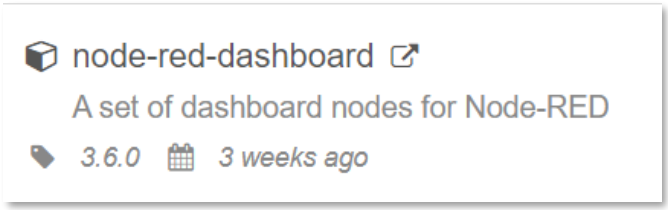
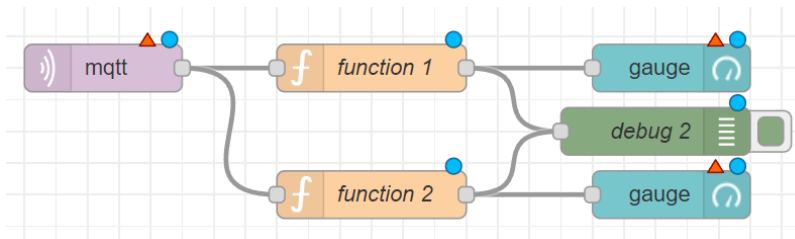
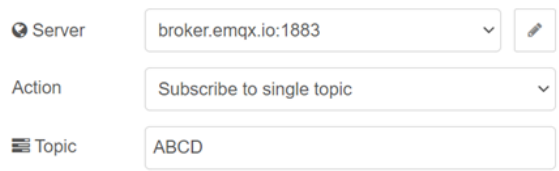
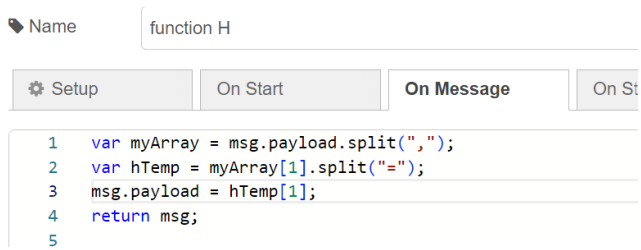
4. Add Dash Board Node

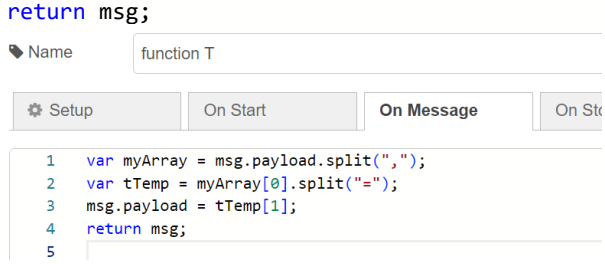
Manage Palates ➔ node-red-dashboard

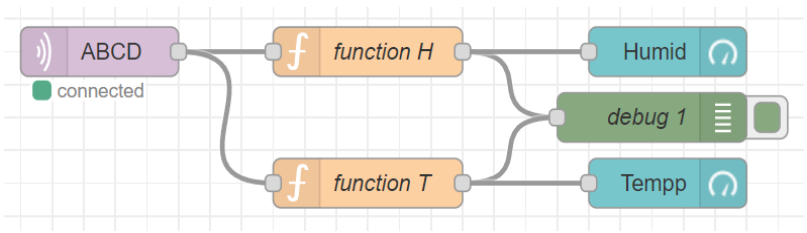


5. Read from Broker and Show on Dashboard

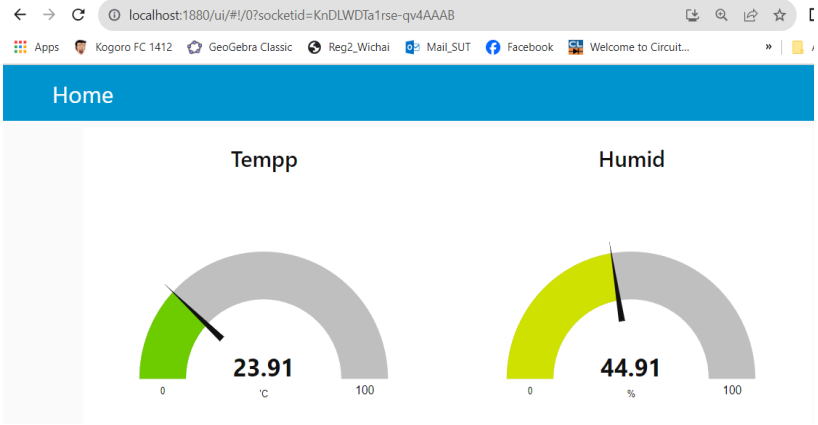
Add Node • Mqtt in • Function 1 • Function 2 • Gauge 1 • Gauge 2 • debug	
Edit mqtt in • Broker = broker.emqx.io:1883 • Topic = ABCD	

Edit Function 1
• Name = function H
	`var myArray = msg.payload.split(",");`
	`var hTemp = myArray[1].split("=");`
	`msg.payload = hTemp[1];`
	`return msg;`
	 

Edit Function 1
• Name = function T
	`var myArray = msg.payload.split(",");`
	`var tTemp = myArray[0].split("=");`
	`msg.payload = tTemp[1];`
	`return msg;`
	 

Edit Gauge 1 • Add & Edit Group	Name: Default Tab: Home Class: Optional CSS class name(s) for widget Width: 12
• Edit Gauge 1	Group: [Home] Default Size: 6 x 6 Type: Gauge Label: Humid Value format: {{value}} Units: % Range: min 0 max 100
Edit Gauge 2 • Edit Gauge 2	Group: [Home] Default Size: 6 x 6 Type: Gauge Label: Tempp Value format: {{value}} Units: 'C Range: min 0 max 100
Deploy	

Dashboard
http://localhost:1880/ui



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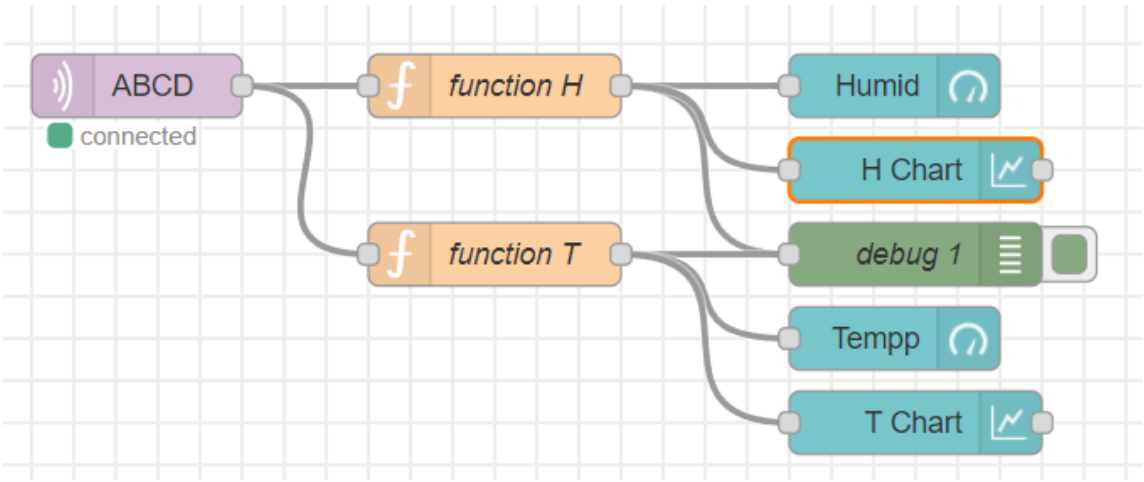
6. Read from Broker and Show on Dashboard – Update



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Neue,sans-serif","edited":true,"reset":false},"darkTheme":{"default":"#097479","baseColor":"#097479","baseFont":"-apple-system,BlinkMacSystemFont,Segoe UI,Roboto,Oxygen-Sans,Ubuntu,Cantarell,Helvetica Neue,sans-serif","edited":false},"customTheme":{"name":"Untitled Theme 1","default":"#4B7930","baseColor":"#4B7930","baseFont":"-apple-system,BlinkMacSystemFont,Segoe UI,Roboto,Oxygen-Sans,Ubuntu,Cantarell,Helvetica 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msg;\n","outputs":1,"timeout":0,"noerr":0,"initialize":"","finalize":"","libs":[],"x":340,"y":120,"wires":[["56643ea8cadd4e29","2c567b34b95e1441","58511e40ee1052e5"]]},{"id":"cd73a2e50dbc11c9","type":"function","z":"d6deba054a936189","name":"function T","func":"var myArray = msg.payload.split(\",\");\nvar tTemp = myArray[0].split(\"=\");\nmsg.payload = tTemp[1];\nreturn 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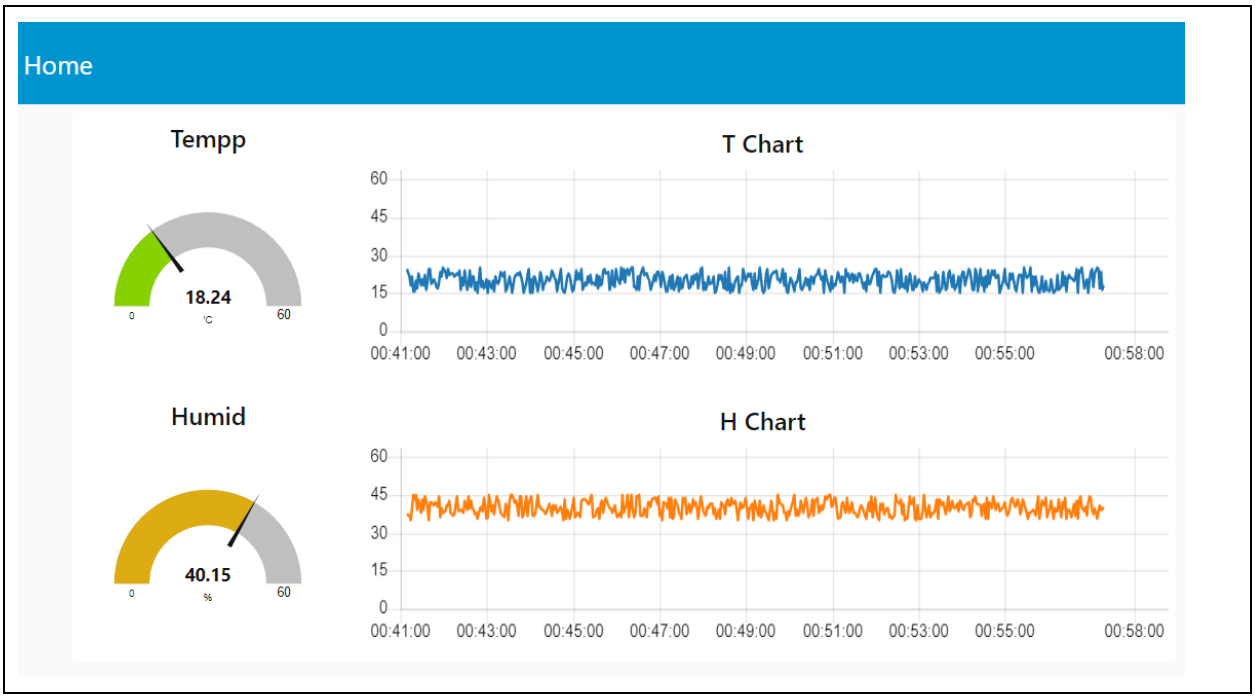
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°C
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T Chart
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H Chart
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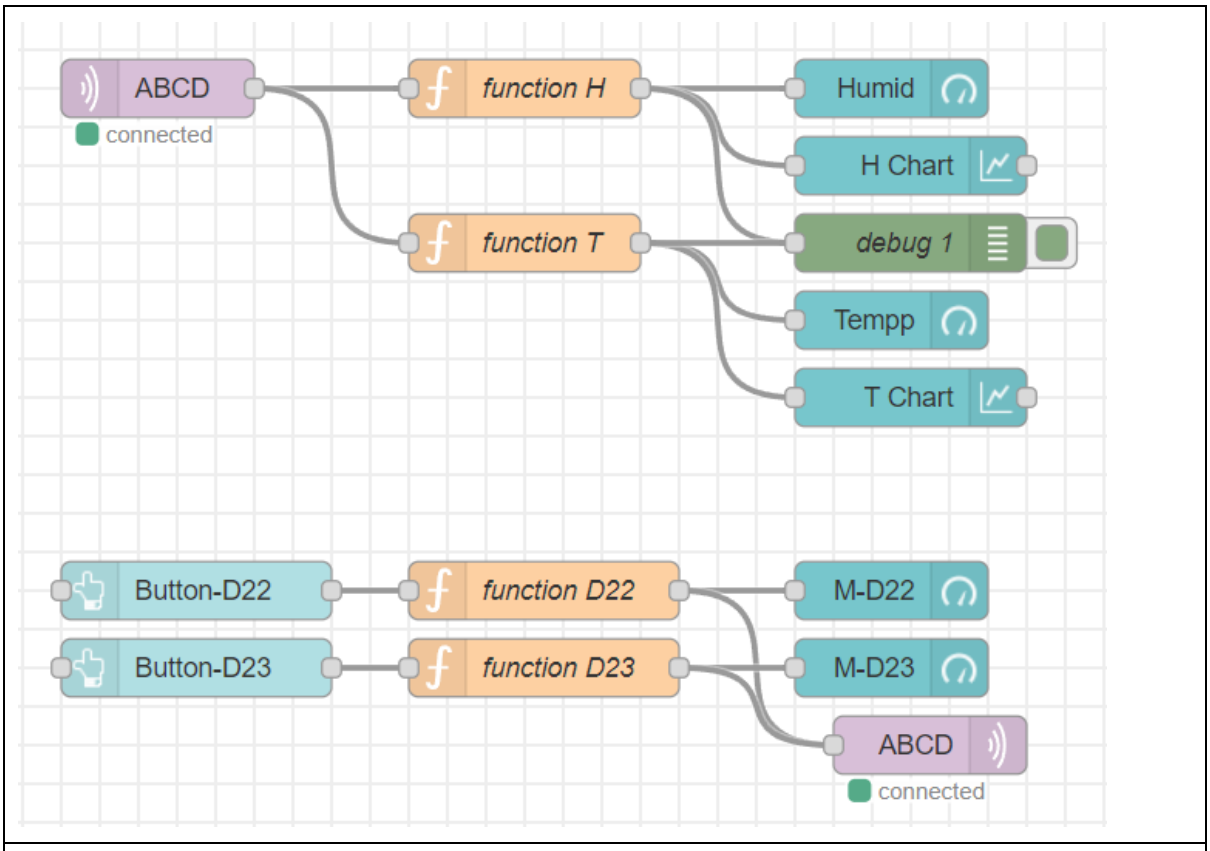
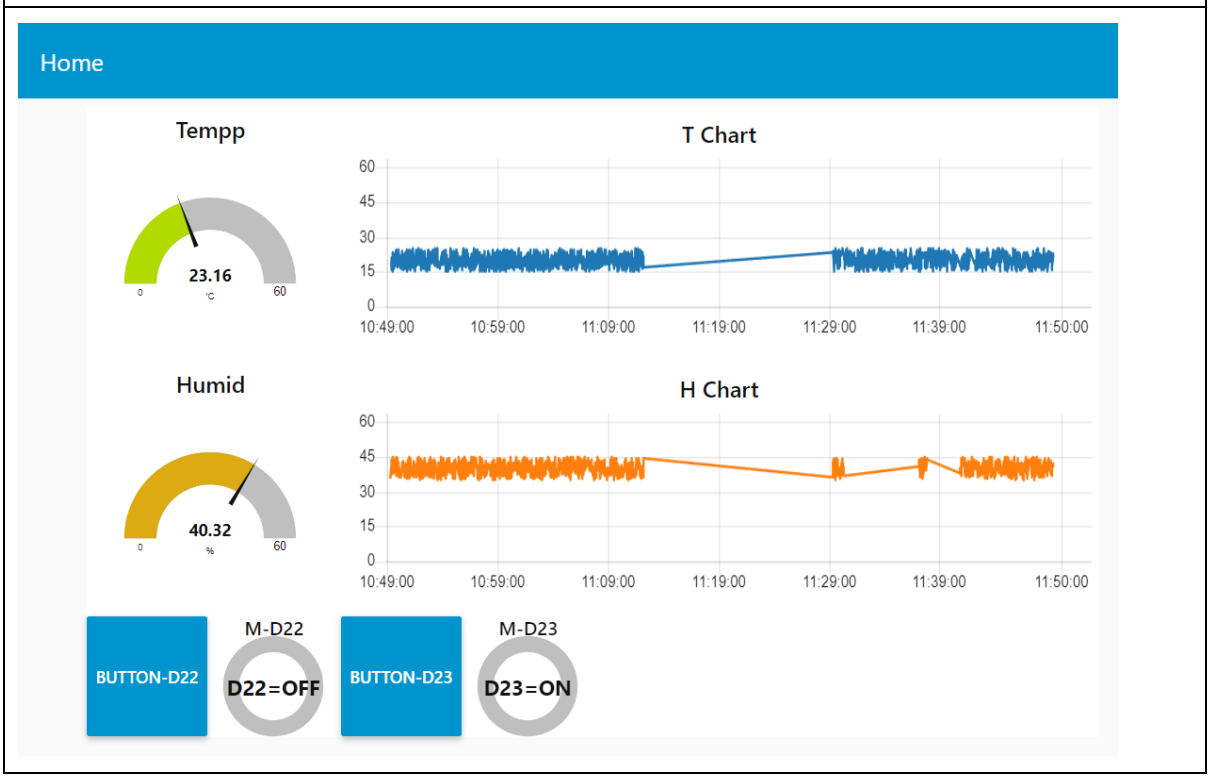
7. Read from Broker, Show and Control from Dashboard





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serif","edited":true,"reset":false},"darkTheme":{"default":"#097479","baseColor":"#097479","baseFont":"-apple-system,BlinkMacSystemFont,Segoe
UI,Roboto,Oxygen-Sans,Ubuntu,Cantarell,Helvetica Neue,sans-serif","edited":false},"customTheme":{"name":"Untitled Theme
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nowHumid)\nmsg.payload = nowHumid\nreturn
msg;\n","outputs":1,"timeout":0,"noerr":0,"initialize":"","finalize":"","libs":[],"x":340,"y":120,"wires":[["2c567b34b95e1441","58511e40ee1052e5","56643
ea8cadd4e29"]]},{"id":"cd73a2e50dbc11c9","type":"function","z":"d6deba054a936189","name":"function T","func":"var nowTempp =
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Add Node-RED Example 1: NodeRED read Oil Price

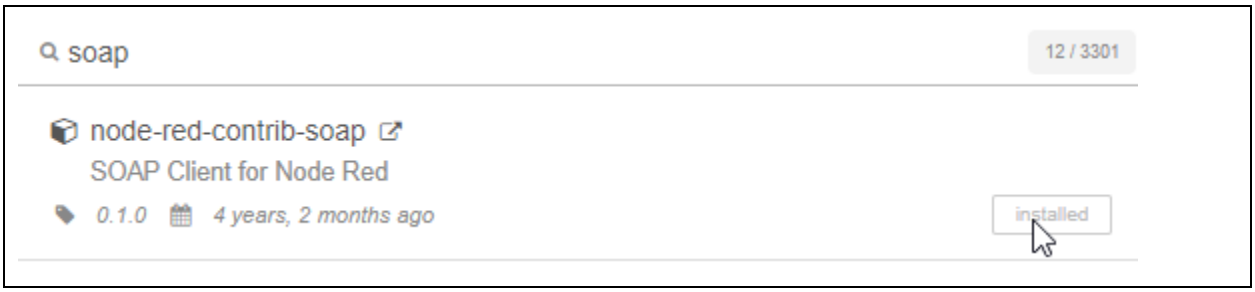
Read

https://medium.com/@shadowxp8rt/ดึงข้อมูลราคาน้ำมันจากเวบ-ปตท-ด้วย-api-ง่ายนิดเดียว-fc462b0053e0

https://webunique.in.th/blog/2017/04/04/การใช้-php-ดึงราคาน้ำมันจา/

https://www1.pttor.com/oilprice-backlink.aspx
https://flows.nodered.org/node/node-red-contrib-soap
https://python3.wannaphong.com/2016/05/python_28.html

Experiment



1. With NodeRED – http request

2. NodeRED – Split JSON to Text

3. Source

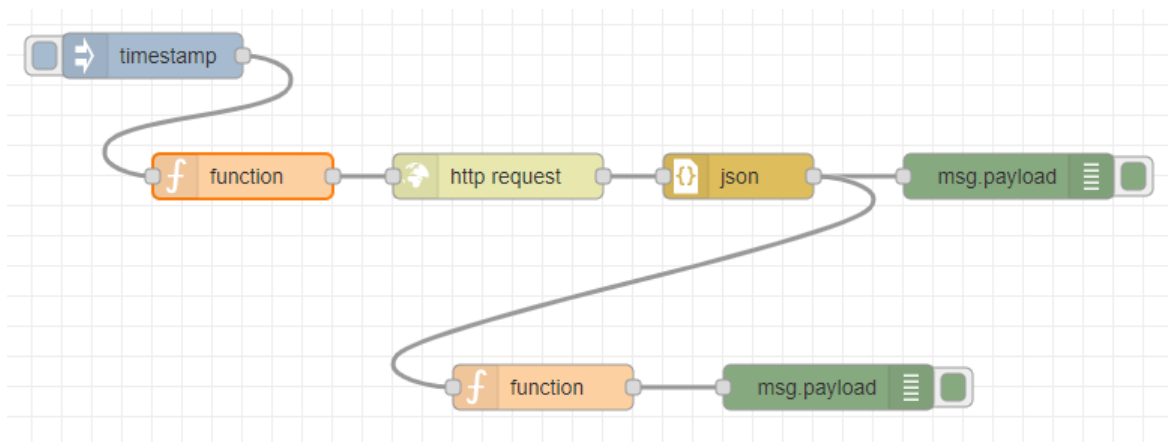
https://www.krupreecha.com/

Wichai Srisuruk

wichai@sut.ac.th

Pk0071212312121

API KEY	**d26d7c40347c257cffb1924d80f1aabc**
SECRET KEY	**55f5aeacd2ae854bd38798743fe20315**



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Add Node-RED Example 2: NodeRED read Anto.io

Read

https://stevesnoderedguide.com/working-with-json-data-node-red

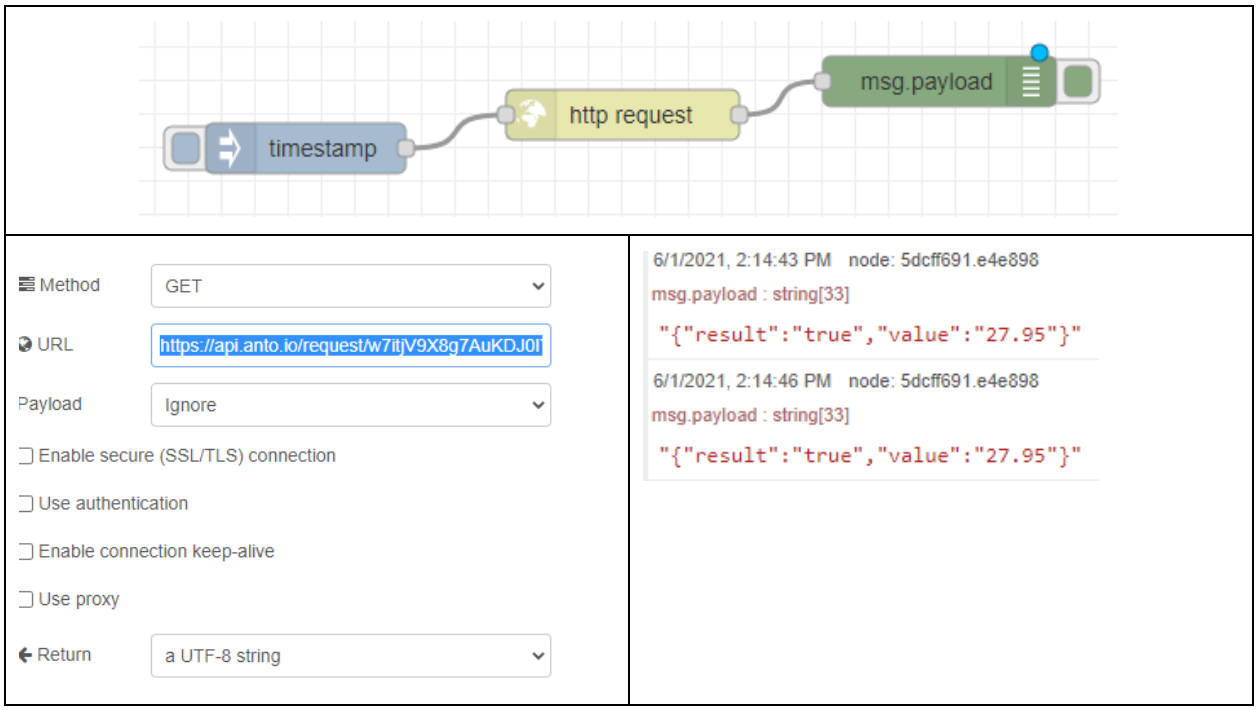
https://antoiot.gitbooks.io/th-document/content/api-and-library/http-api.html

Experiment

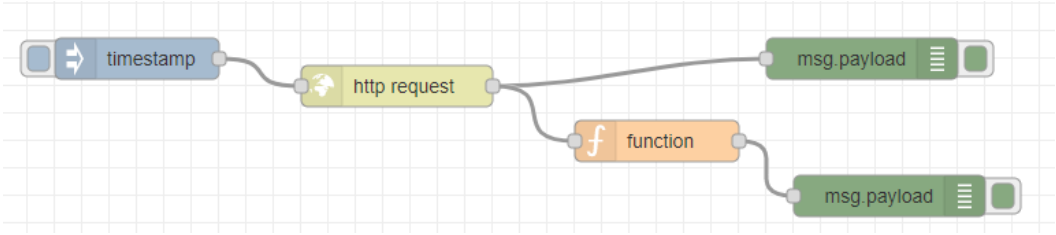
4. Token Key = from ANto.io
5. Test

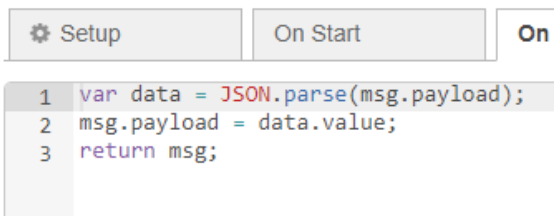
- Request >> **https://api.anto.io/request/TokenKey/gas.95e10.price"**
- For Example >> **https://api.anto.io/request/xxxxxxxxxxxxxxxxxxxxxxxxxxx/gas.95e10.price**

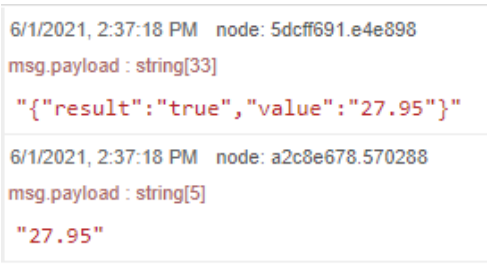
6. With NodeRED – http request



7. NodeRED – Split JSON to Text







8. Source

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Add Node-RED Example 3: NodeRED read lottery data

Read

- https://www.krupreecha.com/
- https://discourse.nodered.org/t/multiple-inputs-in-function/20738
-

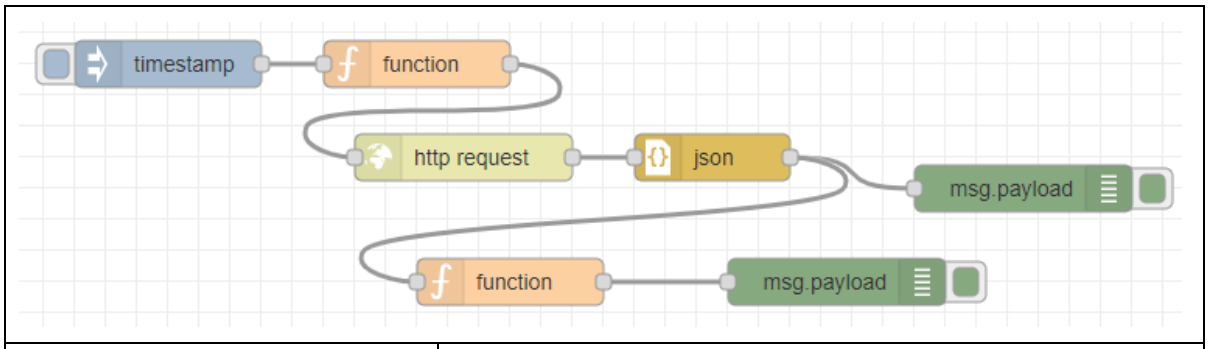
Experiment

1. Get API Key
   https://www.krupreecha.com/
   Name: Wichai Srisuruk
   Email: wichai@sut.ac.th
   Pass: Pk0071212312121
   API KEY: d26d7c40347c257cffb1924d80f1aabc
   SECRET KEY: 55f5aeacd2ae854bd38798743fe20315

2. Using JAVA Scrip

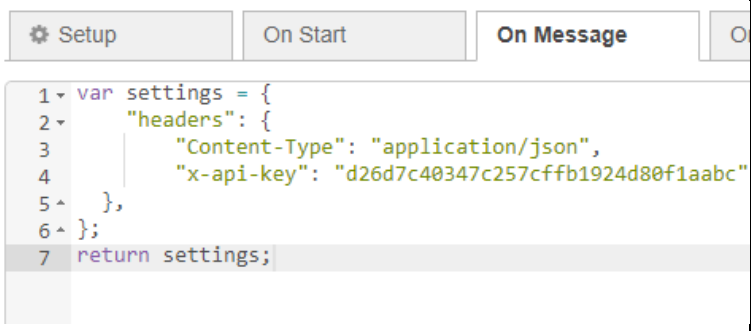
```
// ตัวอย่างการดูหวยงวดวันที่ 16 กุมภาพันธ์ 2564
var settings = {
  "url": "https://api.krupreecha.com/12022564",
  "method": "POST",
  "headers": {
    "Content-Type": "application/json",
    "x-api-key": "api key here",
  },
};

$.ajax(settings).done(function (response) {
  console.log(response);
});
```

3. With Node RED



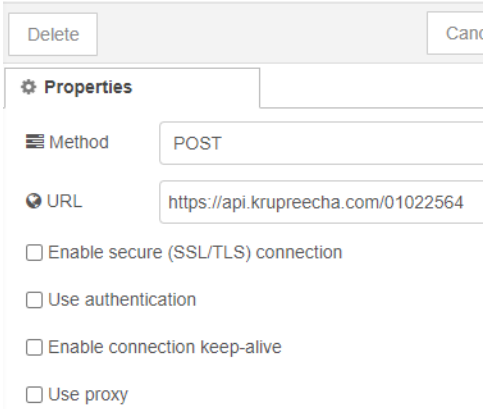
Function

```
var settings = {
   "headers": {
      "Content-Type": "application/json",
      "x-api-key": "your API Key",
 },
};
return settings;
```

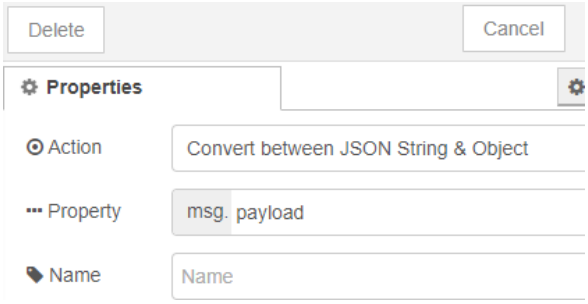


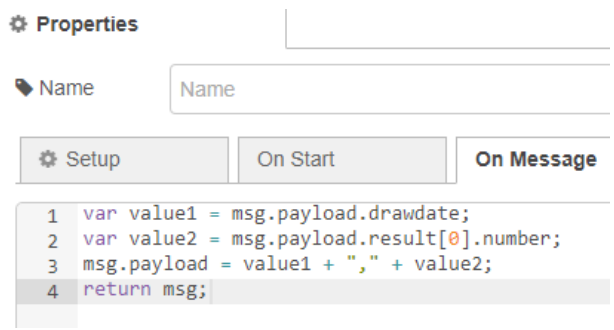
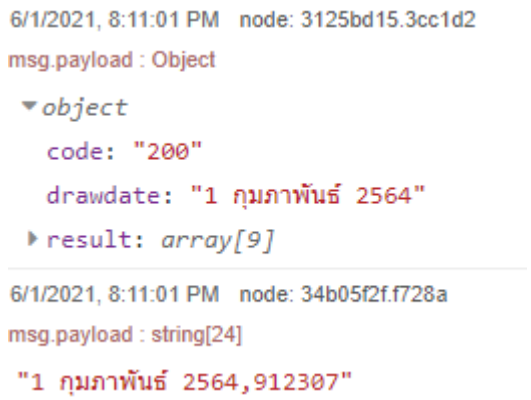
http request

https://api.krupreecha.com/01022564



Json



Function **var value1 = msg.payload.drawdate;** **var value2 = msg.payload.result[0].number;** **msg.payload = value1 + "," + value2;** **return msg;**	
Result	

4. Source

[{"id":"2ab0c5d5.9535ca","type":"tab","label":"Flow 2","disabled":false,"info":""},{"id":"51f697fa.a8ec68","type":"inject","z":"2ab0c5d5.9535ca","name":"","props":[{"p":"payload"},{"p":"topic","vt":"str"}],"repeat":"","crontab":"","once":false,"onceDelay":0.1,"topic":"","payload":"","payloadType":"date","x":120,"y":80,"wires":[["338bcd94.32b922"]]},{"id":"338bcd94.32b922","type":"function","z":"2ab0c5d5.9535ca","name":"","func":"var settings = {\n \"headers\": {\n   \"Content-Type\": \"application/json\",\n   \"x-api-key\": \"d26d7c40347c257cffb1924d80f1aabc\",\n },\n};\nreturn settings;","outputs":1,"noerr":0,"initialize":"","finalize":"","libs":[],"x":280,"y":80,"wires":[["414c98ca.960738"]]},{"id":"414c98ca.960738","type":"http request","z":"2ab0c5d5.9535ca","name":"","method":"POST","ret":"txt","paytoqs":"ignore","url":"https://api.krupreecha.com/01022564","tls":"","persist":false,"proxy":"","authType":"","x":310,"y":140,"wires":[["d0ff2dcc.dd768"]]},{"id":"d0ff2dcc.dd768","type":"json","z":"2ab0c5d5.9535ca","name":"","property":"payload","action":"","pretty":false,"x":470,"y":140,"wires":[["3125bd15.3cc1d2","da43a538.aa9538"]]},{"id":"3125bd15.3cc1d2","type":"debug","z":"2ab0c5d5.9535ca","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"payload","targetType":"msg","statusVal":"","statusType":"auto","x":670,"y":160,"wires":[]},{"id":"da43a538.aa9538","type":"function","z":"2ab0c5d5.9535ca","name":"","func":"var value1 = msg.payload.drawdate;\nvar value2 = msg.payload.result[0].number;\nmsg.payload = value1 + \",\" + value2;\nreturn msg;","outputs":1,"noerr":0,"initialize":"","finalize":"","libs":[],"x":340,"y":220,"wires":[["34b05f2f.f728a"]]},{"id":"34b05f2f.f728a","type":"debug","z":"2ab0c5d5.9535ca","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"false","statusVal":"","statusType":"auto","x":550,"y":220,"wires":[]}]